

Weakness, Singular Learning Theory, and Punctuated Equilibrium in Evolutionary Algorithms and EDAs

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Abstract

We develop a conceptual and technical link between weakness-based schema theorems, singular learning theory (SLT), and the dynamics of evolutionary algorithms (EAs) and estimation of distribution algorithms (EDAs). Weakness is used as a compositional fitness/Occam measure; SLT provides the local learning coefficient (LLC) that quantifies degeneracy of minima via the exponent of local evidence. In the infinite-population limit, we interpret evolutionary dynamics as gradient flows of a free-energy functional, i.e. minimal-free-energy (minimal-weakness) geodesics in population or parameter space. We then show how qualitative “stages” of evolution, including punctuated-equilibrium-like behaviour, can be interpreted as SLT phase changes between different singular strata, each with its own LLC.

We apply this viewpoint to genetic algorithms (GAs) and EDAs on deceptive trap landscapes. For a factorized (univariate) EDA on a simple 2-bit trap, we explicitly compute the free energy and examine the Hessian near deceptive and global optima. We then discuss a 4-bit trap for both factorized and block-factor (dependency-aware) EDAs, explaining how the latter changes the SLT free-energy landscape so that the global basin becomes thermodynamically preferred. This provides a geometric SLT explanation of why dependency-aware EDAs are less susceptible to deception. Finally, we discuss the large-but-finite population case and identify conditions under which infinite-population SLT/weakness conclusions remain qualitatively valid.

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1 Introduction

The aim of this paper is to connect three strands of theory that are usually studied separately:

- *Evolutionary algorithms and EDAs*, which evolve large populations of candidate solutions by selection, mutation, crossover, and probabilistic modeling.
- *Weakness*, a compositional Occam-like evaluation combining fitness and simplicity in a way that interacts well with schema theorems.
- *Singular learning theory* (SLT), which characterizes how Bayesian evidence behaves near singular minima and introduces the local learning coefficient (LLC) as a measure of degeneracy.

At an intuitive level, evolutionary search in a large population is a process that finds and refines families of solutions, not isolated points. There is typically a large amount of redundancy: many distinct genotypes have essentially the same phenotypic effect or fitness. This redundancy is exactly the sort of degeneracy that SLT studies. At the same time, weakness supplies an Occam bias favoring more compressible, more reusable patterns.

Our guiding slogan is:

Infinite-population evolutionary learning of weak models follows minimal-free-energy (minimal-weakness) geodesics, and its developmental stages correspond to SLT phase transitions between singular strata with different LLC, giving a mathematically grounded version of punctuated equilibrium.

We explore this in two concrete settings: a bitstring genetic algorithm (GA) with selection, mutation, and crossover; and simple EDAs in both factorized and dependency-aware forms. On deceptive trap landscapes, we show how the SLT/weakness geometry of the free-energy landscape differs between factorized and block-factor EDAs, and why the latter are structurally better at escaping deception.

We begin with a brief review of evolutionary algorithms, weakness, and SLT, then construct a free-energy picture of evolution, and finally work through deceptive-trap examples for EDAs in some detail.

2 Background

The goal of this section is to spell out the basic ingredients we will use later: evolutionary algorithms and EDAs, the weakness perspective on schema theorems, and the SLT notion of local evidence and LLC.

2.1 Evolutionary algorithms and EDAs

We consider bitstring genotypes $x \in \{0, 1\}^N$. A simple GA evolves a population of N_{pop} individuals over discrete generations:

- *Selection*: choose parents with probability proportional to a fitness (or weakness) score $w(x)$.
- *Crossover*: recombine parent bitstrings using k -point crossover.
- *Mutation*: flip each bit with probability μ .

In the infinite-population limit, the population can be represented by a frequency distribution $p_t(x)$ over genotypes, and the dynamics become deterministic. Under suitable scaling, this can be approximated by a replicator–mutator equation, which we will view as a gradient flow of a free-energy functional.

An estimation-of-distribution algorithm (EDA) maintains and updates a parametric distribution $q_\theta(x)$ over genotypes. At each generation:

1. Sample a population $x^{(1)}, \dots, x^{(M)}$ from q_θ .
2. Select individuals based on $w(x)$ (e.g. top- α or stochastic selection).
3. Fit a new parameter θ' (e.g. maximum likelihood) to the selected set.

In the infinite-sample limit, the θ -update becomes a deterministic mapping that we can describe as gradient descent on a suitable free-energy functional.

We will use two EDA families:

- *Factorized EDA*: assumes independent bits, $q_\theta(x) = \prod_{i=1}^N \theta_i^{x_i} (1 - \theta_i)^{1-x_i}$.
- *Block-factor EDA*: groups bits into blocks b and models each block distribution $q_b(x_b)$, so $q_\Theta(x) = \prod_b q_b(x_b)$.

2.2 Weakness and schema theorems

Weakness is a way of assigning a real-valued cost or “badness” to a program or genotype that is compatible with compositional structures and schema theorems. For this paper we do not need the full categorical formalism; it suffices to note that we can:

- treat $w(x)$ as a weakness score, lower is better or higher is better depending on convention;
- interpret schemata as coarse descriptors of bit patterns;
- and reason about expected schema frequencies over generations under selection, mutation, and crossover.

Classical schema theorems state that certain schemata with above-average fitness proliferate exponentially in expectation, if crossover and mutation do not disrupt them too much. When weakness is used as the fitness signal, similar results hold: schemata representing more compressible, more reusable patterns are favoured.

From the SLT perspective, the size of a schema class (how many concrete genotypes instantiate it) is a discrete proxy for the *volume* of near-optimal parameters. In the continuous analysis that follows, we will replace schema counts with actual volume integrals in parameter space.

2.3 Singular learning theory and local learning coefficient

SLT focuses on the behavior of Bayesian evidence in singular models. Given a loss (or negative log-likelihood) $\ell(\theta)$, a prior density $\phi(\theta)$, and n data points, the (local) evidence integral is

$$Z_n = \int \exp(-n\ell_n(\theta)) \phi(\theta) d\theta,$$

where ℓ_n is the empirical loss. Near a minimizer θ^* , and under analyticity assumptions, SLT shows that

$$-\log Z_n = n\ell_n(\theta^*) + \lambda(\theta^*) \log n + O(\log \log n). \quad (1)$$

The coefficient $\lambda(\theta^*)$ is called the *local learning coefficient* (LLC). It plays the role of an effective model complexity:

- in regular models of dimension d , $\lambda = d/2$;
- in singular models, λ may be fractional and is determined by the singularity structure (e.g. manifolds of minima, symmetries).

Equivalently, one can look at sublevel sets of the excess loss $K(\theta) = \ell(\theta) - \ell(\theta^*) \geq 0$. Define the local volume

$$V_K(\varepsilon) = \int_{\{K(\theta) \leq \varepsilon\}} \phi(\theta) d\theta.$$

Then there exist $\lambda > 0$ and $m \in \mathbb{N}$ such that

$$V_K(\varepsilon) = \varepsilon^\lambda (\log(1/\varepsilon))^{m-1} (c + o(1)), \quad (\varepsilon \downarrow 0), \quad (2)$$

so λ describes how quickly the volume of near-optimal parameters shrinks with tolerance.

In this paper we use LLC qualitatively: lower λ means a fatter basin with more degeneracy; higher λ means a stiffer, less degenerate minimum.

3 Weakness and Evolution as Free-Energy Geodesics

Before diving into SLT details, it is helpful to adopt a free-energy viewpoint on evolution. This is a standard trick in evolutionary dynamics, but here we connect it explicitly to weakness.

3.1 Infinite-population GA as a gradient flow

Let $p_t(x)$ be the population distribution over genotypes at generation t in the infinite-population limit. Under selection and mutation (and with a simple recombination model), one can often write the dynamics as

$$\partial_t p_t(x) = \mathcal{S}[p_t](x) + \mathcal{M}[p_t](x) + \mathcal{R}[p_t](x), \quad (3)$$

where $\mathcal{S}$, $\mathcal{M}$, and $\mathcal{R}$ are selection, mutation, and recombination operators respectively. In the weak-mutation/weak-selection limit, this can be approximated by a Fokker–Planck or replicator–mutator PDE.

To connect this to weakness, we define:

- a potential (energy) $V(x) = -\log w(x)$, so high weakness corresponds to low energy (if w is large when fitness is good);
- a free-energy functional on distributions p :

$$F[p] = \int V(x)p(x) dx + \beta^{-1} \int p(x) \log p(x) dx,$$

where the first term is expected energy and the second term is an entropy-like regularizer weighted by β^{-1} .

Under appropriate choices of metric (e.g. Wasserstein-2 or a replicator geometry), the evolution of p_t can be written as a gradient flow

$$\partial_t p_t = -\nabla_G F[p_t], \tag{4}$$

i.e. the population distribution follows the steepest descent of free energy. We do not need the metric details here; the key idea is that:

Infinite-population evolution follows minimal-free-energy geodesics in the space of probability distributions.

3.2 Weakness schema theorem as a discrete SLT shadow

In a weakness-based schema theorem, one tracks counts $m_s(t)$ of individuals matching a schema s , and shows that under selection and modest mutation and crossover, schemata with above-average weakness score w_s proliferate exponentially in expectation. Schematically,

$$m_s(t+1) \approx m_s(t) \frac{w_s}{\bar{w}(t)} (1 - \text{disruption}),$$

where $\bar{w}(t)$ is the population-average weakness and the disruption term depends on mutation and crossover.

From the SLT viewpoint, the class of genotypes matching schema s has a certain size $|s|$ (number of concrete genotypes). The fact that evolution concentrates on schemata with large w_s and large $|s|$ is a discrete analogue of concentrating posterior mass on regions of parameter space that have both low loss and large volume. Passing from discrete schemas to continuous parameter volumes is a natural step if we wish to apply SLT to evolutionary attractors.

4 SLT for Evolutionary Attractors and Punctuated Equilibrium

In this section we interpret evolutionary attractors as minima of a free-energy functional and qualitatively import SLT’s notion of LLC to them. We then connect evolutionary stage changes to phase flips between basins.

4.1 Population-as-parameter SLT

Instead of treating θ as parameters, we can treat the population distribution p itself as an unknown element of a manifold $\mathcal{P}$, e.g. the simplex of distributions over a finite space. The free-energy functional $F[p]$ plays the role of loss.

In principle, one can apply SLT to $F[p]$ just as to $F(\theta)$: consider a parameterization of $\mathcal{P}$ (for instance, via sufficient statistics) and analyse the local volume of distributions near a minimizer p^* ,

$$\{p : F[p] \leq F[p^*] + \varepsilon\}.$$

If the minimizer is regular and isolated, the local geometry is quadratic and the corresponding LLC is $(d/2)$, where d is the dimension of the parameterization. If there are neutral directions (continuous families of equally good distributions), the LLC is smaller, reflecting degeneracy.

Thus, in an evo-dynamics context, one can interpret:

- lower LLC at p^* as a broader, more redundant attractor, with many almost-equivalent population states;
- higher LLC as a stiffer attractor with fewer neutral directions.

4.2 Stage transitions as SLT phase flips

Suppose there are multiple local minima p_i^* of $F[p]$, corresponding to different evolutionary “macro-organizations” (different ecotypes, modes, or modular structures). In SLT, the contribution of each minimum to evidence involves both the free energy and the LLC. If one had an effective sample size n (for example, interpreting n as the amount of fitness information or the length of an evolutionary episode), one could write a schematic expansion

$$F_i(n) \approx nF[p_i^*] + \lambda_i \log n.$$

As n increases, the tradeoff between the leading term nF_i and the subleading $\lambda_i \log n$ can cause the identity of the most thermodynamically favourable basin to change. At moderate scales, a slightly worse free-energy minimum with very low LLC (fat basin) may dominate; at large scales, a slightly better free-energy minimum with higher LLC may take over.

In evolutionary terms, this is reminiscent of *punctuated equilibrium*: long periods where the population fluctuates near one attractor, followed by relatively rapid transitions to a different attractor when the balance of fit vs degeneracy changes due to environmental change or internal structural discovery. The SLT language clarifies this as a phase flip between singular strata.

5 Factorized EDA on Deceptive Traps: A Concrete Analysis

We now step into a concrete setting where we can write formulas explicitly: a univariate (factorized) EDA on a deceptive trap landscape. We first work out a 2-bit case in detail and then discuss the 4-bit trap qualitatively.

5.1 Univariate EDA on a 2-bit deceptive trap

We consider bitstrings $x = (x_1, x_2) \in \{0, 1\}^2$ and a simple deceptive trap function. Define the trap via the number of ones $u(x) = x_1 + x_2$:

$$f(u) = \begin{cases} 2 & \text{if } u = 2, \\ 2 - 1 - u & \text{if } u = 0, 1. \end{cases}$$

Thus

$$f(00) = 1, \quad f(01) = 0, \quad f(10) = 0, \quad f(11) = 2.$$

We take $w(x) := f(x)$ as the selection weight.

EDA model and evidence. A univariate EDA assumes an independent-bit model

$$q_\theta(x_1, x_2) = \theta_1^{x_1} (1 - \theta_1)^{1-x_1} \theta_2^{x_2} (1 - \theta_2)^{1-x_2},$$

with $\theta = (\theta_1, \theta_2) \in (0, 1)^2$. The selection-reweighted evidence is

$$Z(\theta) = \sum_{x \in \{0,1\}^2} w(x) q_\theta(x).$$

Enumerating:

$$\begin{aligned} q(00) &= (1 - \theta_1)(1 - \theta_2), \\ q(01) &= (1 - \theta_1)\theta_2, \\ q(10) &= \theta_1(1 - \theta_2), \\ q(11) &= \theta_1\theta_2, \end{aligned}$$

so

$$\begin{aligned} Z(\theta) &= 1 \cdot q(00) + 0 \cdot q(01) + 0 \cdot q(10) + 2 \cdot q(11) \\ &= (1 - \theta_1)(1 - \theta_2) + 2\theta_1\theta_2 \\ &= 1 - \theta_1 - \theta_2 + 3\theta_1\theta_2. \end{aligned}$$

We define the free energy

$$F(\theta) := -\log Z(\theta).$$

Global optimum and deceptive corner. On the domain $[0, 1]^2$, critical points and optima of F correspond to critical points and extrema of $Z(\theta)$. At the corners:

$$\begin{aligned} Z(0, 0) &= 1, \\ Z(1, 0) &= 1 - 1 - 0 + 0 = 0, \\ Z(0, 1) &= 1 - 0 - 1 + 0 = 0, \\ Z(1, 1) &= 1 - 1 - 1 + 3 = 2. \end{aligned}$$

Thus:

- $(\theta_1, \theta_2) = (1, 1)$ is the global maximizer of Z , hence global minimizer of F ;
- $(0, 0)$ is better than $(1, 0)$ and $(0, 1)$ but not globally optimal.

The global optimum $(1, 1)$ corresponds to the true global genotype optimum 11. The corner $(0, 0)$ is a local attractor for many EDA dynamics because each bit, considered in isolation, sees that setting it to 1 tends to reduce fitness, due to the deceptive shape of f .

Hessian analysis. To study degeneracy at $(1, 1)$ and $(0, 0)$, we compute the gradient and Hessian of F .

We have

$$Z(\theta) = 1 - \theta_1 - \theta_2 + 3\theta_1\theta_2, \quad \nabla Z(\theta) = \begin{bmatrix} -1 + 3\theta_2 \\ -1 + 3\theta_1 \end{bmatrix}, \quad \nabla^2 Z(\theta) = \begin{bmatrix} 0 & 3 \\ 3 & 0 \end{bmatrix}.$$

Using $F = -\log Z$ and the chain rule,

$$\begin{aligned} \nabla F(\theta) &= -\frac{1}{Z(\theta)} \nabla Z(\theta), \\ \nabla^2 F(\theta) &= \frac{1}{Z(\theta)^2} \nabla Z(\theta) \nabla Z(\theta)^\top - \frac{1}{Z(\theta)} \nabla^2 Z(\theta). \end{aligned}$$

At the global optimum $(1, 1)$:

$$Z(1, 1) = 2, \quad \nabla Z(1, 1) = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \quad \nabla^2 Z(1, 1) = \begin{bmatrix} 0 & 3 \\ 3 & 0 \end{bmatrix}.$$

Thus

$$\nabla^2 F(1, 1) = \frac{1}{4} \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 0 & 3 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 - 1.5 \\ 1 - 1.5 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix}.$$

The eigenvalues of this symmetric matrix are 1 ± 0.5 , i.e. 1.5 and 0.5, both positive. So F is locally strictly convex at $(1, 1)$: a regular, non-degenerate minimum.

At $(0, 0)$:

$$Z(0, 0) = 1, \quad \nabla Z(0, 0) = \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \quad \nabla^2 Z(0, 0) = \begin{bmatrix} 0 & 3 \\ 3 & 0 \end{bmatrix},$$

so

$$\nabla^2 F(0, 0) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 3 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix},$$

with eigenvalues 3 and -1 . Thus $(0, 0)$ is a saddle point in the unconstrained continuous domain $\mathbb{R}^2$, with one stable and one unstable direction.

However, in the actual EDA parameter domain $[0, 1]^2$ with clipping at the boundaries, the unstable direction may be effectively obstructed: once both θ_i are near 0, selection tends to keep them near zero, and mutation / noise may be too weak to push them back into the interior. From a dynamical standpoint, $(0, 0)$ behaves like a deceptive attractor.

SLT/weakness interpretation. In this tiny 2-bit example, the global basin around $(1, 1)$ is regular and non-singular; the local geometry is well captured by the Hessian. If we consider many independent 2-bit traps concatenated into a higher-dimensional problem, we get:

- A global basin where each pair's (θ_1, θ_2) block is near $(1, 1)$. This basin is relatively small in parameter volume because each coordinate is tightly constrained.
- A deceptive basin where each pair's parameters are near $(0, 0)$. Because this is a corner of the domain, many nearby (θ_1, θ_2) configurations are nearly as bad, and the combinatorial number of such configurations grows rapidly with the number of blocks.

On such multi-block problems, the deceptive basin can have a much larger volume (and lower LLC) than the global basin, making it thermodynamically attractive in SLT terms despite worse expected fitness. This is a free-energy/degeneracy restatement of the familiar fact that univariate EDAs can get stuck in deceptive traps.

5.2 Factorized EDA on a 4-bit trap (qualitative)

For a 4-bit deceptive trap, each block $x_b = (x_{b,1}, \dots, x_{b,4})$ has fitness $f_b(u)$ defined in terms of $u = \sum_i x_{b,i}$, with a high reward at $u = 4$ and deceptively higher values near $u = 0$ than near intermediate values. Global fitness is a sum over blocks.

A univariate EDA still assumes independent bits, with parameters $\theta \in (0, 1)^N$. The free energy

$$F(\theta) = -\log \sum_x f(x) q_\theta(x)$$

now has the following features:

- The model class cannot represent block-level correlations. This smears the true elite distributions across many bit patterns.
- There is a broad deceptive basin in θ -space where each block's bits have marginals biased toward the deceptive patterns.
- The global basin near correct 4-bit patterns (each block's bits near 1) exists but is narrower: many small changes in marginals reduce the chance of sampling the exact optimal pattern.

From an SLT/weakness perspective, the deceptive basin is supported by combinatorial degeneracy in θ -space: many different marginal parameter settings give similar behaviour. In higher dimensions, this translates into smaller LLC (fatter basin), and the SLT free-energy expansion $F(n) \approx nF^* + \lambda \log n$ may favour the deceptive basin up to large n .

6 Dependency-Aware EDAs: Phase Flip in the SLT Landscape

We now consider a dependency-aware EDA that can model 4-bit blocks directly, and explain how this changes the SLT free-energy geometry on the same deceptive trap landscape.

6.1 Block-factor EDA on a 4-bit trap

Suppose we group bits into 4-bit blocks b , and define an EDA model

$$q_\Theta(x) = \prod_b q_b(x_b),$$

where each q_b is a full distribution on $\{0, 1\}^4$ (or an equivalent small Bayesian network). For the 4-bit trap, this model can represent the exact elite distributions for each block: high mass on $(1, 1, 1, 1)$ and low mass on other patterns.

The free energy in this parameter space is

$$F(\Theta) = -\log \sum_x f(x) q_\Theta(x).$$

Qualitatively, this landscape has:

- A global basin where each block q_b is concentrated near the true optimal configuration.
- Much weaker deceptive basins: because the model can distinguish between different (4-bit) patterns, it does not need to smear probability mass across deceptive and non-deceptive genotypes.

6.2 SLT/weakness perspective: a phase flip

In the factorized EDA, the deceptive basin was supported by strong degeneracy in parameter space: many different marginal patterns approximate the deceptive attractor. The global basin was less degenerate.

In the block-factor EDA:

- The global basin becomes thermodynamically dominant: the block model can place probability mass on the true optimum with little spillover, lowering the free energy F substantially.
- The deceptive configurations no longer benefit from the same degree of degeneracy: the model can explicitly represent and penalize them.

This is naturally described as a *phase flip* in the SLT landscape: when the model class is enriched from independent bits to block factors, the local minima and their LLCs change in such a way that the global basin overtakes the deceptive basin.

From an evolutionary stage perspective:

- In the early factorized regime, EDAs behave similarly to univariate models, with stage-like behaviour dominated by deceptive basins.
- Once enough structure is observed to warrant block factors (structure learning), the landscape changes qualitatively, and the EDA enters a new stage where convergence to the global optimum is much more robust.

7 Finite Population Effects: When Do Infinite-Pop Conclusions Hold?

So far we have mostly spoken about infinite populations and infinite sample sizes. In practical evolutionary algorithms and EDAs, populations are finite. Here we outline how finite-population noise interacts with the infinite-pop SLT/weakness picture.

7.1 Noise scaling in finite populations

Consider a bitwise marginal θ_i estimated from a population or selected subpopulation of size M . Then

$$\hat{\theta}_i = \frac{1}{M} \sum_{j=1}^M x_i^{(j)}$$

has variance $\text{Var}(\hat{\theta}_i) = \theta_i(1 - \theta_i)/M$. Thus the estimation noise per coordinate is of order $1/\sqrt{M}$.

In both GA and EDA:

- Infinite-pop (or infinite-sample) dynamics correspond to a deterministic update $\theta \mapsto \theta'$ or $p \mapsto p'$.
- Finite-pop dynamics add a noise term of order $1/\sqrt{N_{\text{pop}}}$ per coordinate per generation.

Over time, this noise can accumulate and cause the population to drift between basins of attraction.

7.2 Selection vs drift and SLT basins

A classical rule of thumb in population genetics is that selection dominates drift when $Ns \gg 1$, where s is a selection coefficient and N is population size. The SLT/weakness version can be expressed in terms of:

- free-energy difference between basins, ΔF ;
- curvature (or more generally, local Lipschitz) of F around minima;
- noise scale $\sigma^2 \sim 1/N$ in the relevant coarse-grained coordinates.

Qualitatively, the infinite-pop SLT picture (geodesics and phase flips between basins) is a good qualitative guide when:

$$N \Delta F_{\text{stage}} \gg 1, \quad \frac{1}{\sqrt{N}} \ll \text{typical basin scale},$$

so that random fluctuations are too small to constantly push the system across SLT basin boundaries.

In this regime, finite populations spend most of their time near the same attractors as in the infinite-pop limit, and the stage structure predicted by the SLT/weakness analysis remains qualitatively valid.

8 Discussion

We have argued that:

- Evolutionary algorithms and EDAs can be understood as gradient flows of free-energy functionals, with weakness as the “energy” term.
- Singular learning theory provides a way to quantify degeneracy of evolutionary attractors via the local learning coefficient, λ .
- Stage-like behaviour (often described as punctuated equilibrium) can be viewed as SLT phase changes between basins with different free-energy and degeneracy profiles.
- On deceptive trap landscapes, factorized EDAs possess a free-energy landscape where deceptive basins are thermodynamically favoured over substantial ranges, whereas dependency-aware EDAs change the SLT geometry so that the global basin is structurally favoured.

Conceptually, this suggests that:

- Redundancy and neutral directions in large evolving populations are not just “noise”, but an essential part of the geometry of learning.
- Designing evolutionary algorithms and EDAs with richer dependence models can be seen as changing the singularity structure (and LLCs) so that deceptive basins lose their advantage.
- SLT/weakness tools could, in principle, be used to predict when a given evolutionary system is likely to undergo a qualitative stage transition, and to tune algorithm parameters to exploit or avoid such transitions.

9 Conclusion

We have outlined a connection between weakness-based schema theorems, singular learning theory, and the dynamics of evolutionary algorithms and EDAs. Infinite-population GA and EDA dynamics can be viewed as minimal-free-energy geodesics in population or parameter space, with SLT LLCs quantifying the degeneracy of attractors. On deceptive landscapes, factorized EDAs and dependency-aware EDAs differ sharply in their SLT/weakness geometry, providing a principled explanation of the practical observation that dependency-aware EDAs handle deception more robustly.

We view this as a first step toward a more systematic use of SLT and weakness in the analysis and design of evolutionary learners, and as a bridge between ideas from statistical learning theory, evolutionary computation, and the study of punctuated equilibrium in complex adaptive systems.

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